

SAI NITHIN P

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PROFESSIONAL EXPERIENCE

WALMART

Sunnyvale, CA

Software Engineer III

Aug 2024 - Feb 2026

- Owned and shipped Android Cart and Checkout features end-to-end (concepting, design, implementation, release, and maintenance), modeling UI state to survive process death, configuration changes, and low-memory conditions without corrupting cart or payment state.
- Applied mastery of Kotlin with structured concurrency — Coroutines, Flow, and StateFlow (reactive streams, RxJava-style operators) — to coordinate asynchronous cart, pricing, and payment updates and prevent race conditions.
- Architected lifecycle-safe navigation with ViewModel-scoped state and explicit UI state machines (MVVM/MVI), using Dagger/Hilt dependency injection to keep complex, multi-team flows maintainable.
- Built UI with Jetpack Compose and integrated GraphQL (Apollo) and REST APIs over Retrofit/OkHttp, shaping queries to minimize payload, stabilize pagination, and tolerate partial backend failures.
- Worked at the application-engine and SDK level with ExoPlayer and MediaCodec (adaptive bitrate, buffering, audio focus, lifecycle-bound cleanup) and native interop (JNI/C++) for performance-critical paths.
- Drove performance and maintainability by profiling with the Android Studio Profiler; prototyped new frameworks and applied functional, immutable patterns to improve productivity and developer experience.
- Improved reliability with ViewModel and UI tests in Kotlin (JUnit, Espresso, Mockito) inside Git-based CI workflows, design reviews, and technical documentation.

CVS HEALTH

Arlington, VA

Android Developer

Aug 2023 - Jul 2024

- Owned Android Prescription Refill and Order Placement flows in Kotlin, modeling durable UI state to survive process death, configuration changes, and low-memory conditions without duplicating refill requests or corrupting order state.
- Coordinated asynchronous workflows with Coroutines, Flow, and reactive patterns, modeling durable order state separately from transient UI events to prevent duplicate submissions and inconsistent refill status.
- Integrated REST APIs (Retrofit/OkHttp) for eligibility, pricing, insurance validation, and order confirmation, validating response schemas and error contracts under latency or partial failures.
- Implemented Android authentication using OAuth 2.0 and JWT, storing tokens in EncryptedSharedPreferences and handling refresh and expiration to keep sessions secure.
- Structured features into clean data, domain, and presentation layers with Dagger/Hilt dependency injection, improving testability and reducing coupling across flows.
- Optimized RecyclerView and screen rendering for prescription lists and order summaries by stabilizing item IDs and minimizing layout invalidations, delivering smoother scrolling and faster UI response.

COGNIZANT TECHNOLOGY SOLUTIONS

Hyderabad, India

Java Developer

Jun 2019 - Aug 2021

- Implemented Java services with Spring Boot and added authentication and authorization via Spring Security, improving system reliability and reducing error rates in healthcare applications.
- Designed and implemented a continuous-integration pipeline using Maven and Jenkins, with Git-based branching workflows, code reviews, and pull requests to maintain code quality.
- Deployed Java microservices in Docker containers orchestrated by Kubernetes for faster rollout and higher availability, automating build, test, and delivery.
- Integrated Apache Kafka for event-driven architectures, enabling real-time data streaming across microservices.
- Designed RESTful Spring Boot microservice APIs with input validation, pagination, and centralized exception handling for reliable, well-documented service contracts.
- Implemented Redis caching for frequently accessed data, reducing read latency and offloading the database under high traffic.
- Wrote unit and integration tests with JUnit and Mockito, raising coverage and catching regressions early in CI.
- Analyzed database performance and implemented indexing strategies, reducing database latency by 20%.

PROJECTS

- Transaction-Safe Android Checkout System:** Built a lifecycle-safe Android checkout flow in Kotlin using state-driven MVVM and Coroutines/Flow to handle process death, background execution limits, and idempotent backend interactions without duplicate submissions.
- Low-Latency Adaptive Video Streaming Engine:** Implemented an Android video player using ExoPlayer with HLS/DASH, tuning MediaCodec decoding, buffering, and adaptive-bitrate logic to reduce startup latency and rebuffering under unstable networks.
- Full Stack Web Application - "Yummy" Recipe Platform:** Developed a recipe discovery and sharing site with a responsive front-end (HTML, CSS, Bootstrap, JavaScript, jQuery) backed by Oracle DB. mason.gmu.edu/~spore2/Yummy

SKILLS

Programming Languages: Kotlin, Java, C, C++, C#, Go

Android & Jetpack: Activities, Fragments, Jetpack Compose, ViewModel, LiveData/StateFlow, Coroutines/Flow, Room, DataStore, WorkManager, MVVM, MVI, state-driven UI, Lifecycle-aware Architecture, Dagger/Hilt, ExoPlayer, JNI

Networking & APIs: REST, Apollo GraphQL, Retrofit, OkHttp, JSON, pagination, partial-failure handling, reactive streams (Flow / RxJava-style)

Testing & Performance: JUnit, Espresso, Mockito, UI/instrumentation testing, TDD, Android Profiler, Memory Analysis, Jank diagnosis

Security & Auth: OAuth 2.0, JWT, EncryptedSharedPreferences, Spring Security

Backend, Messaging & DevOps: Spring Boot, Microservices, Apache Kafka, Redis, RTSP/WebRTC, Docker, Kubernetes, Maven, Jenkins, Git, CI/CD, Agile, Scrum, database indexing & query optimization

CERTIFICATIONS

- Google Generative AI; GCP - Professional Machine Learning Engineer; AWS Cloud Practitioner; HackerRank - Data Structures (Advanced), Problem Solving (Advanced), Python Programming; Oracle - Java Programming; EPAM - Introduction to Front-end Technology, Database & Testing

EDUCATION

George Mason University | Master of Science, Computer Science

Fairfax, VA, USA